

FITBUDDY AI

Phase 2: Requirement Analysis

Prepared from the current public FitBuddy_ai repository and organized according to the Phase 1–6 structure of the provided AI/ML & GenAI project template.

2.1 Functional Requirements

ID	Requirement
FR-01	The system shall provide a home page with a profile registration form.
FR-02	The form shall collect name, age, gender, weight, height, workout days, objective and experience level.
FR-03	The backend shall accept the form through the /generate-plan endpoint.
FR-04	The system shall create and store an original workout plan.
FR-05	The system shall display the registered profile and active plan on the dashboard.
FR-06	The system shall support viewing an individual user by ID.
FR-07	The system shall accept feedback through the /submit-feedback endpoint.
FR-08	The system shall store updated_plan and feedback_text in SQLite.
FR-09	The system shall provide an all-users administrative view.

2.2 Non-Functional Requirements

- Usability: the application should be understandable to a first-time user.
- Maintainability: backend routes, database logic and templates should remain separated.
- Reliability: invalid user IDs should return a clear HTTP 404 response.
- Performance: normal form submission and SQLite reads/writes should complete quickly for a small project dataset.
- Security: API keys should be kept in environment configuration rather than hard-coded in source.
- Extensibility: the architecture should allow a future Gemini generation layer without redesigning the entire UI.

2.3 Software Requirements

Component	Requirement
Language	Python
Backend	FastAPI
Templates	Jinja2 HTML templates
Database	SQLite
Configuration	Pydantic Settings / .env
Development environment	VS Code and Python virtual environment
Version control	Git / GitHub

2.4 Data Requirements

The visible database implementation creates a users table with an auto-incrementing ID, profile attributes, original_plan, updated_plan and feedback_text.

2.5 User Input Requirements

- Name — required text field.
- Age — required numeric field.
- Gender — selectable profile value.
- Weight — required numeric value in kilograms.
- Height — required numeric value in centimetres.
- Workout days — selectable 3, 4 or 5 day options in the current UI.
- Objective — weight loss, muscle gain, endurance or general health in the current UI.
- Experience level — beginner, intermediate or advanced in the current UI.

2.6 Requirement Traceability

Phase 2 requirements map directly to the current repository components: main.py initializes FastAPI, routes.py handles the web workflow, database.py creates and accesses SQLite, and the HTML templates provide the user dashboard and administrator registry.

Project repository: https://github.com/asvikamoorthy-cyber/FitBuddy_ai

Reference structure: <https://github.com/Ravi-teja-777/AI-ML-and-GEN-AI-Track-Project-Template>

Scope note: This document describes the project state visible in the public repository and identifies proposed/future items separately where applicable.